

IGCSE Past Paper Exam: Question 1:

- 1 The following questions all use even parity. Identify the parity bit required and add it to the end of each question.

a 101011 b 010000 c 0101001
 d 11111111 e 01100101 f 010101010100

- 2 The following questions all use odd parity. Identify the parity bit required and add it to the end of each question.

a 010101 b 000011 c 111100
 d 11000011 e 001111100 f 010101010

Parity checking: IGCSE Exam: Question 2

- (d) When eight bytes of data have been collected, they are transmitted to a computer 100km away. Parity checks are carried out to identify if the data has been transmitted correctly. The system uses **even parity** and column 1 is the parity bit.

The eight bytes of data are sent together with a ninth parity byte:

	parity bit	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	1	0	0	0	0	1	0	0
byte 2	1	1	1	1	0	0	1	1
byte 3	0	1	0	0	1	0	0	0
byte 4	0	1	1	1	0	0	0	1
byte 5	1	0	0	0	1	1	1	1
byte 6	0	0	0	0	0	0	0	0
byte 7	1	1	1	0	1	0	0	0
byte 8	1	0	0	0	1	1	1	0
parity byte	1	0	1	1	0	1	1	1

- (i) Identify which of the eight bytes contains an error.

byte[1]

IGCSE Exam: Question 2

- (ii) Identify which column contains an error.

column[1]

- (iii) The incorrect bit is indicated where the byte number and column cross.

Give the corrected byte.

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[1]

- (iv) Calculate the denary value of the corrected byte.

.....
[1]

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3

IGCSE Exam: Question 3

- (a) Parity checks are often used to detect errors that may occur during data transmission.

The received bytes in the table below were transmitted using **odd parity**.

Tick (✓) to show whether each byte has been **corrupted during transmission** or **not corrupted during transmission**.

Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)
10110100		
01101101		
10000001		

[3]

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4

IGCSE Exam: Question 4

- 8 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

- (a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01100100	
10010001	
00000011	
10110010	

[4]

- (b) An error may **not** be detected when using a parity check.

Identify why an error may **not** be detected.

.....
 [1]

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5

IGCSE Exam: Question 4

- (c) The data is sent using parallel half-duplex data transmission.

- (i) Describe how data is sent using parallel half-duplex data transmission.

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 [4]

- (ii) State **two** drawbacks of using parallel data transmission.

Drawback 1

 Drawback 2
 [2]

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6

IGCSE Exam: Question 5

- 6 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

- (a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01111100	
10010000	
10011001	
00101001	

[4]

- (b) The 8-bit binary value 10110001 is transmitted and received as 10110010

A parity check does **not** identify any errors in the binary value received.

State why the parity check does **not** identify any errors.

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.....

[1]

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7

IGCSE Exam: Question 5

- (c) The data is sent using serial duplex data transmission.

- (i) Describe how data is sent using serial duplex data transmission.

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[4]

- (ii) State **one** drawback of using serial data transmission, rather than parallel data transmission.

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[1]

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8

IGCSE Exam: Question 6

- 9 Four 7-bit binary values are being transmitted from one computer to another. An odd parity check is being used to check for errors in the binary values.

Write the correct **Parity bit** for each **7-bit binary value** to make sure it meets **odd** parity.

Parity bit	7-bit binary value
.....	0000011
.....	1000000
.....	0111111
.....	1010101

[4]

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9

IGCSE Exam: Question 7

- 9 The contents of three binary registers have been transmitted from one computer to another. **Even parity** has been used as an error detection method.

The outcome after transmission is:

Register A and **Register C** have been transmitted **correctly**.

Register B has been transmitted **incorrectly**.

Complete the **Parity bit** for each register to show the given outcome.

	Parity bit							
Register A		0	1	0	0	1	0	1
Register B		1	0	0	0	0	0	1
Register C		1	0	0	0	0	1	1

[3]

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10

IGCSE Exam: Question 8

- 5 The three binary numbers in the registers given have been transmitted from one computer to another.

One binary number has been transmitted incorrectly. This can be identified by the use of a **Parity bit**.

Identify the binary number that has been transmitted **incorrectly**. Explain how you identified the incorrect binary number.

	Parity bit							
Register A	1	0	1	1	1	0	0	1
Register B	1	1	1	0	0	1	1	1
Register C	1	0	0	1	1	0	1	1

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11

IGCSE Exam: Question 8

The binary number that has been transmitted incorrectly is in **Register**

Explanation

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 [4]

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12

Automatic Repeat Request (ARQ): IGCSE Exam: Question 9

- (b)** Another method of error detection is Automatic Repeat reQuest (ARQ).

Explain how ARQ is used in error detection.

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..... [4]

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13

Automatic Repeat Request (ARQ): IGCSE Exam: Question 10

- 2** Parity checks and Automatic Repeat reQuests (ARQ) can be used to check for errors during data transmission and storage.

- (a)** A system uses **even parity**. Write the appropriate parity bit for each byte.

Parity Bit							
	1	0	1	0	0	1	1
	1	0	1	1	1	1	1
	1	0	1	0	0	0	1

[2]

- (b)** Explain how Automatic Repeat reQuests (ARQ) are used in data transmission and storage.

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.....

..... [2]

- (c)** State **one** other method that could be used to check for transmission errors.

..... [1]

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14

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Question 11

- 7 Umar is a manager in an office. The data that he uses on a daily basis is stored on a file server close to his computer.

Data is sent from his computer to the file server using parallel simplex data transmission.

- (a) Describe how the data is sent using parallel simplex data transmission.

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..... [3]

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15

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Question 11

- (b) The file server is moved to another building that is 1 km away.

Explain why the parallel simplex data transmission method that Umar uses is no longer suitable.

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..... [2]

- (c) Checksum and Automatic Repeat reQuest (ARQ) are both used when transmitting the data from a computer to the file server.

Explain why checksum and ARQ are both used.

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..... [3]

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16

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

- 1 A company selling CDs uses a unique 6-digit identification number for each CD title. The right-most digit (position 1) is a *check digit*.

For example,

6	5	4	3	2	1	← digit position
3	0	6	1	4	9	← identification number
					↑	
					check digit	

The validity of the number and check digit is calculated as follows:

- ☐ multiply **each** digit by its digit position
- ☐ add up the results of the multiplications
- ☐ divide the answer by 11
- ☐ if the remainder is 0, the identification number and check digit are valid.

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17

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

- (a) Show whether the following identification numbers are valid or not. You **must** show how you arrived at your answer.

Identification number 1: 4 2 1 9 2 3

working:

.....

.....

valid or not valid?

Identification number 2: 8 2 0 1 5 6

working:

.....

.....

valid or not valid? [3]

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18

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 12

(b) Find the check digit for this identification number.

5 0 2 4 1 ____

working:

.....

.....

check digit: [2]

(c) Describe, with examples, **two** different types of data entry errors that a check digit would detect.

1

.....

2

..... [2]

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19

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 13

There are various methods used to detect errors that can occur during data transmission and storage.

Describe each of the following error detection methods.

Parity check

.....

.....

Check digit

.....

.....

Checksum

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Automatic Repeat request (ARQ)

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.....

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[8]⁰

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 14

Identify and describe **two** methods of error checking that can be used to make sure that the data stored after transmission is accurate.

Method 1

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Method 2

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[6]

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21

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Question 15

Five computer terms and seven descriptions are shown below.

Draw a line to connect each computer term to its correct description.

Computer term	Description
Serial, simplex data transmission	Several bits of data sent down several wires, in both directions, but not at the same time
Parallel, half-duplex data transmission	Several bits of data sent down several wires, in both directions, at the same time
Parity check	An even or odd number of bits set to 1 in a byte, used to check if the byte has been transmitted correctly
Automatic repeat request (ARQ)	One bit sent at a time, over a single wire in one direction only
Checksum	An additional digit placed at the end of a number to check if the number has been entered correctly
	A value transmitted at the end of a block of data; it is calculated using the other elements in the data stream and is used to check for transmission errors
	An error detection method that uses response and time out when transmitting data; if a response is not sent back to the sender in an agreed amount of time, then the data is re-sent

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[5]

22

IGCSE Exam: Question 16

- 9 (a) Computers can transmit data using different methods.

Describe the **three** data transmission methods given.

(i) Serial data transmission

.....

 [2]

(ii) Parallel data transmission

.....

 [2]

(iii) Duplex data transmission

.....

 [2]

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23

IGCSE Exam: Question 16

- (b) Data can sometimes be corrupted when it is transmitted from one computer to another, causing errors to be present in the data.

Identify and describe **three** methods of error detection that could be used to see if an error has occurred.

Error detection method 1

Description

.....

Error detection method 2

Description

.....

Error detection method 3

Description

.....

[9]

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24

IGCSE Exam: Question 17

- 5 The three binary numbers in the registers X, Y and Z have been transmitted from one computer to another.

								Parity bit
Register X	1	0	0	1	0	0	1	0
Register Y	1	1	1	0	0	1	1	1
Register Z	1	1	1	0	1	0	0	1

Only **one** binary number has been transmitted correctly. This is identified through the use of a parity bit.

Identify which register contains the binary number that has been transmitted **correctly**. Explain the reason for your choice.

The binary number that has been transmitted correctly is in **Register**

Explanation

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25

IGCSE Exam: Question 18

- 4 A company transmits data to external storage at the end of each day.

- (a) Parity checks can be used to check for errors during data transmission.

The system uses **odd parity**.

- (i) Tick (✓) to show for each of the received bytes whether they have been **transmitted correctly** or **transmitted incorrectly**.

Received byte	Transmitted correctly (✓)	Transmitted incorrectly (✓)
10001011		
10101110		
01011101		
00100101		

[4]

- (ii) State **one** other method that could be used to check for transmission errors.

..... [1]

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26

IGCSE Exam: Question 18

(b) Data can be transferred using parallel or serial data transmission.

(i) Describe what is meant by parallel data transmission.

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.....
.....
..... [2]

(ii) Give **one** application of parallel data transmission.

.....
..... [1]

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27

IGCSE Exam: Question 19

5 (a) Parity checks are often used to detect errors that may occur during data transmission.

The received bytes in the table below were transmitted using **odd parity**.

Tick (✓) to show whether each byte has been **corrupted during transmission** or **not corrupted during transmission**.

Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)
10110100		
01101101		
10000001		

[3]

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28

IGCSE Exam: Question 19

- (b) Another method of error detection is Automatic Repeat reQuest (ARQ).

Explain how ARQ is used in error detection.

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..... [4]

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29

IGCSE Exam: Question 20

- 9 Check digits are used to ensure the accuracy of entered data.

A 7-digit number has an extra digit on the right, called the check digit.

digit position:	1	2	3	4	5	6	7	8
digit:	—	—	—	—	—	—	—	—
								↑ check digit

The check digit is calculated as follows:

- each digit in the number is multiplied by its digit position
- the seven results are then added together
- this total is divided by 11
- the remainder gives the check digit (if the remainder = 10, the check digit is X)

- (a) Calculate the check digit for the following number. Show all your working.

4 2 4 1 5 0 8 ...

.....

.....

.....

Check digit [2]

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30

IGCSE Exam: Question 20

(b) An operator has just keyed in the following number:

3 2 4 0 0 4 5 X

Circle below **correct** if the check digit is correct **OR incorrect** if the check digit is incorrect.

correct incorrect

Explain your answer.

.....

.....

.....

.....

[3]

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31

IGCSE Exam: Question 21

- 4 Nine bytes of data are transmitted from one computer to another. Even parity is used. An additional parity byte is also sent.

The ten bytes arrive at the destination computer as follows:

	parity bit	bit 2	bit 3	bit 4	bit 5	bit 6	bit 7	bit 8
byte 1	1	1	1	0	1	1	1	0
byte 2	0	0	0	0	0	1	0	1
byte 3	0	1	1	1	1	0	0	0
byte 4	1	1	0	0	0	0	0	0
byte 5	1	0	1	1	1	1	1	0
byte 6	0	1	0	1	1	0	0	1
byte 7	0	1	1	1	0	0	1	1
byte 8	0	0	1	1	0	1	1	0
byte 9	1	1	0	0	0	0	1	1
parity byte	0	0	1	0	0	0	1	0

One of the bits was corrupted during the data transmission.

- (a) **Circle** the corrupt bit in the corrupt byte in the table above.

[1]

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32

IGCSE Exam: Question 21

- (b) Explain how the corrupted bit was found.

.....

.....

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.....

.....

.....[2]

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33

IGCSE Exam: Question 22

- 9 In the following barcode, each binary number is made up of seven bars.

Each bar is black or grey.

A black bar is interpreted as a "1" and a grey bar is interpreted as a "0".

- (a) Write the binary numbers that would be produced from this barcode:



Binary number A Binary number B

Binary number A:

--	--	--	--	--	--	--

Binary number B:

--	--	--	--	--	--	--

[2]

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34

IGCSE Exam: Question 22

(b) This barcode system uses odd parity.

Write the parity bit for each of the binary numbers in **part (a)**:

Binary number A:

Parity bit

Binary number B:

[2]

IGCSE Exam: Question 23

5 Parity checks are often used to check for errors that may occur during data transmission.

(a) A system uses **even parity**.

Tick (✓) to show whether the following three bytes have been transmitted correctly or incorrectly.

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		
0 1 1 1 1 1 0 0		
0 1 1 0 1 0 0 1		

[3]

IGCSE Exam: Question 23

- (b) A parity byte is used to identify which bit has been transmitted incorrectly in a block of data.

The word "F L O W C H A R T" was transmitted using nine bytes of data (one byte per character). A tenth byte, the parity byte, was also transmitted.

The following block of data shows all ten bytes received after transmission. The system uses **even parity** and column 1 is the parity bit.

	letter	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	F	1	0	1	0	0	1	1	0
byte 2	L	1	0	1	0	1	1	0	0
byte 3	O	1	0	1	0	1	1	1	1
byte 4	W	1	0	1	1	0	1	1	1
byte 5	C	1	0	1	0	0	0	1	1
byte 6	H	0	0	1	0	1	0	0	0
byte 7	A	0	0	1	0	0	1	0	1
byte 8	R	1	0	1	1	0	0	1	0
byte 9	T	1	0	1	1	0	1	0	0
parity byte		1	0	1	1	1	1	1	0

- (i) One of the bits has been transmitted incorrectly.

Write the byte number and column number of this bit:

Byte number

Column number [2]

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37

IGCSE Exam: Question 23

- (ii) Explain how you arrived at your answer for part (b)(i).

.....

 [2]

- (c) Give the denary (base 10) value of the byte: **1 0 1 1 1 1 1 0**

.....
 [1]

- (d) A parity check may not identify that a bit has been transmitted incorrectly.

Describe **one** situation in which this could occur.

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 [1]

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38

IGCSE Exam: Question 24

12 Parity checks are used to check for errors during data transmission. A system uses **odd** parity.

- (a) Complete the following two bytes of data so that they both have **odd** parity:

	1	1	1	1	0	0	0
	0	0	0	0	1	1	1

[2]

- (b) Name and describe another method which can be used to check whether data has been correctly transmitted.

Name of method

Description

.....

.....

[2]

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39

IGCSE Exam: Question 25

- 7 (a) Check digits are used to ensure the accuracy of input data.

A 7-digit code number has an extra digit on the right, called the check digit.

Digit position	1	2	3	4	5	6	7	8
Digit	–	–	–	–	–	–	–	–

The check digit is calculated as follows:

- each digit in the number is multiplied by its digit position
- the seven results are then added together
- this total is divided by 11
- the remainder gives the check digit (if the remainder = 10, the check digit is X)

- (i) Calculate the check digit for the following code number. Show all your working.

4 2 4 1 5 0 8 ...

.....

.....

.....

Check digit

[2]

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40

IGCSE Exam: Question 25

(ii) An operator has just keyed in the following code number:

3 2 4 0 0 4 5 X

Has the operator correctly keyed in the code number?

.....

Give a reason for your answer.

.....

.....

.....

[3]

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41

IGCSE Exam: Question 25

(b) When data are transmitted from one device to another, a parity check is often carried out on each byte of data. The parity bit is often the leftmost bit in the byte.

(i) If a system uses even parity, give the parity bit for each of the following bytes:

parity bit

	1	1	0	0	1	1	0
--	---	---	---	---	---	---	---

parity bit

	0	0	0	0	0	0	1
--	---	---	---	---	---	---	---

[2]

(ii) A parity check can often detect corruption of a byte.

Describe a situation in which it **cannot** detect corruption of a byte.

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.....

.....[1]

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42